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Method for manufacturing oscillator

Abstract

A method for manufacturing oscillators of a plurality of types including a first oscillator and a second oscillator, the method includes: manufacturing the first oscillator, the manufacturing the first oscillator including a first-oscillator first step of mounting, to a first container, a first resonator element and a first circuit element configured to oscillate the first resonator element to generate a first oscillation signal, and a first-oscillator second step of mounting, to the first container, a second resonator element whose oscillation frequency is controlled based on the first oscillation signal; and manufacturing the second oscillator, the manufacturing the second oscillator including a second-oscillator first step of mounting a third resonator element and a second circuit element to a second container of the same type as the first container, the third resonator element being of the same type as the first resonator element and the second circuit element being of the same type as the first circuit element and being configured to oscillate the third resonator element to generate a second oscillation signal, and a second-oscillator second step of mounting, to the second container, a fourth resonator element whose oscillation frequency is controlled based on the second oscillation signal and whose frequency is different from the frequency of the second resonator element.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-154774, filed Sep. 28, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a method for manufacturing an oscillator.

2. Related Art

(3) An oven-controlled crystal oscillator (OCXO) disclosed in JP-A-2020-120159 includes a first container accommodating a first resonator element and a first circuit element configured to oscillate the first resonator element to generate a first oscillation signal. A second container accommodating a second resonator element whose oscillation frequency is controlled based on the first oscillation signal is mounted at a lower surface of the first container.

(4) As such an oven-controlled crystal oscillator, it is required to efficiently manufacture a plurality of types of oscillators, to which the second resonator element having a different characteristic is mounted, at low cost.

SUMMARY

(5) A method for manufacturing an oscillator according to the present disclosure is a method for manufacturing oscillators of a plurality of types including a first oscillator and a second oscillator, the method including: manufacturing the first oscillator, the manufacturing the first oscillator including a first-oscillator first step of mounting, to a first container, a first resonator element and a first circuit element configured to oscillate the first resonator element to generate a first oscillation signal, and a first-oscillator second step of mounting, to the first container, a second resonator element whose oscillation frequency is controlled based on the first oscillation signal; and manufacturing the second oscillator, the manufacturing the second oscillator including a second-oscillator first step of mounting a third resonator element and a second circuit element to a second container of the same type as the first container, the third resonator element being of the same type as the first resonator element and the second circuit element being of the same type as the first circuit element and being configured to oscillate the third resonator element to generate a second oscillation signal, and a second-oscillator second step of mounting, to the second container, a fourth based on the second oscillation signal and whose frequency is different from the frequency of the second resonator element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view illustrating a common package according to a first embodiment.

(2) FIG. 2 is a plan view of the common package as viewed from above.

(3) FIG. 3 is a cross-sectional view illustrating an inner package of the common package and an inner portion thereof.

(4) FIG. 4 is a plan view of the inner package as viewed from below.

(5) FIG. 5 is a circuit diagram illustrating a PLL circuit provided in a circuit element of the common package.

(6) FIG. 6 is a circuit diagram illustrating a first route of the PLL circuit.

(7) FIG. 7 is a circuit diagram illustrating a second route of the PLL circuit.

(8) FIG. 8 is a circuit diagram illustrating a third route of the PLL circuit.

(9) FIG. 9 is a cross-sectional view illustrating a first oscillator.

(10) FIG. 10 is a cross-sectional view illustrating a voltage-controlled crystal oscillator provided in the first oscillator.

(11) FIG. 11 is a plan view of the voltage-controlled crystal oscillator as viewed from below.

(12) FIG. 12 is a cross-sectional view illustrating a second oscillator.

(13) FIG. 13 is a cross-sectional view illustrating a third oscillator.

(14) FIG. 14 is a cross-sectional view illustrating a fourth oscillator.

(15) FIG. 15 is a flowchart illustrating a method for manufacturing oscillators.

(16) FIG. 16 is a flowchart illustrating a method for manufacturing oscillators.

(17) FIG. 17 is a cross-sectional view illustrating a common package according to a second embodiment.

(18) FIG. 18 is a cross-sectional view illustrating a common package according to a third embodiment.

DESCRIPTION OF EMBODIMENTS

(19) Hereinafter, preferred embodiments of a method for manufacturing an oscillator according to the present disclosure will be described in detail with reference to the accompanying drawings. For convenience of description, an X-axis, a Y-axis, and a Z-axis orthogonal to one another are illustrated in the drawings. Hereinafter, a direction along the X-axis is also referred to as an “X-axis direction”, a direction along the Y-axis is also referred to as a “Y-axis direction”, and a direction along the Z-axis is also referred to as a “Z-axis direction”. A side indicated by an arrow of each axis is also referred to as a “positive side”, and an opposite side thereof is also referred to as a “negative side”. The positive side in the Z-axis direction is also referred to as “upper”. The negative side in the Z-axis direction is also referred to as “lower”. Further, a plan view from the Z-axis direction is also simply referred to as a “plan view”.

First Embodiment

(20) FIG. 1 is a cross-sectional view illustrating a common package according to a first embodiment. FIG. 2 is a plan view of the common package as viewed from above. FIG. 3 is a cross-sectional view illustrating an inner package of the common package and an inner portion thereof. FIG. 4 is a plan view of the inner package as viewed from below. FIG. 5 is a circuit diagram illustrating a PLL circuit provided in a circuit element of the common package. FIG. 6 is a circuit diagram illustrating a first route of the PLL circuit. FIG. 7 is a circuit diagram illustrating a second route of the PLL circuit. FIG. 8 is a circuit diagram illustrating a third route of the PLL circuit. FIG. 9 is a cross-sectional view illustrating a first oscillator. FIG. 10 is a cross-sectional view illustrating a voltage-controlled crystal oscillator provided in the first oscillator. FIG. 11 is a plan view of the voltage-controlled crystal oscillator as viewed from below. FIG. 12 is a cross-sectional view illustrating a second oscillator. FIG. 13 is a cross-sectional view illustrating a third oscillator. FIG. 14 is a cross-sectional view illustrating a fourth oscillator. FIGS. 15 and 16 are flowcharts each illustrating a method for manufacturing an oscillator.

(21) In a method for manufacturing an oscillator 1 according to the embodiment, a plurality of types of oscillators 1 are manufactured using a common package 100. Hereinafter, as an example, a case where four types of oscillators 1 in total, that is, a first oscillator 1A, a second oscillator 1B, a third oscillator 1C, and a fourth oscillator 1D that are oven-controlled crystal oscillators (OCXO) are manufactured will be described. Thus, by using the common package 100, a plurality of types of oscillators 1 can be manufactured at low cost and with high efficiency. Therefore, first, the common package 100 will be described.

(22) The common package 100 illustrated in FIGS. 1 to 4 is suitably used for an oven controlled crystal oscillator (OCXO), and includes a resonator element 6, a circuit element 8, a temperature control element 7, an inner package 3, a circuit element 4, and an outer package 2. The inner package 3 accommodates the resonator element 6, the circuit element 8, and the temperature control element 7, and the outer package 2 accommodates the inner package 3 and the circuit element 4.

(23) The resonator element 6 corresponds to a first resonator element of the first oscillator 1A, a third resonator element of the second oscillator 1B, and a fifth resonator element of the third oscillator 1C. Accordingly, this means that resonator elements 6 of the same type are used as the first resonator element, the third resonator element, and the fifth resonator element. The resonator elements 6 of the same type mean that the resonator elements 6 have the same resonance frequency, outer shape, shape of each portion represented by an electrode shape, dimension of each portion,

and material (component material) of each portion. In addition, the resonator elements **6** of the same type may mean resonator elements having the same model number or identification code.

(24) The circuit element **8** corresponds to a first circuit element of the first oscillator **1A**, a second circuit element of the second oscillator **1B**, and a fifth circuit element of the third oscillator **1C**. Accordingly, this means that circuit elements **8** of the same type are used as the first circuit element, the second circuit element, and the fifth circuit element. The circuit elements **8** of the same type mean that the circuit elements **8** have the same shape, dimension, circuit configuration, circuit layout, and the like. In addition, the circuit elements **8** of the same type may mean circuit elements having the same model number or identification code.

(25) The temperature control element **7** corresponds to a first heater circuit of the first oscillator **1A** and a second heater circuit of the second oscillator **1B**. Accordingly, this means that temperature control elements **7** of the same type are used as the first heater circuit and the second heater circuit. The temperature control elements **7** of the same type mean that the temperature control elements **7** have the same shape, dimension, circuit configuration, circuit layout, and the like. In addition, the temperature control elements **7** of the same type may mean temperature control elements having the same model number or identification code.

(26) Outer Package **2**

(27) As illustrated in FIG. **1**, the outer package **2** includes an outer base **21** and an outer lid **22**. The outer base **21** has a box shape, and includes an upper recessed portion **211** opening in an upper surface thereof and a lower recessed portion **212** opening in a lower surface thereof. The outer lid **22** has a plate shape, and is bonded to the upper surface of the outer base **21** via a sealing member **23** such as a seal ring or low-melting glass so as to close an opening of the upper recessed portion **211**. Accordingly, the upper recessed portion **211** is hermetically sealed, and an accommodating space **Q2** is formed in the outer package **2**. On the other hand, an opening of the lower recessed portion **212** is not sealed and faces an outside of the outer package **2**.

(28) Component materials of the outer base **21** and the outer lid **22** are not particularly limited. For example, the outer base **21** may be made of various ceramic materials containing alumina, titania or the like, and the outer lid **22** may be made of various metal materials such as Kovar.

(29) The accommodating space **Q2** will be described in detail, the upper recessed portion **211** includes a first upper recessed portion **211a** that is open in the upper surface, a second upper recessed portion **211b** that is open in a bottom surface of the first upper recessed portion **211a** and has an opening smaller than that of the first upper recessed portion **211a**, and a third upper recessed portion **211c** that is open in a bottom surface of the second upper recessed portion **211b** and has an opening smaller than that of the second upper recessed portion **211b**. The circuit element **4** is disposed at the bottom surface of the first upper recessed portion **211a**, and the inner package **3** is disposed at a bottom surface of the third upper recessed portion **211c**.

(30) The accommodating space **Q2** is airtight and is in a depressurized state, preferably in a state close to vacuum. Accordingly, thermal insulation of the outer package **2** is enhanced. In addition, heat of the temperature control element **7** is less likely to escape, and heating efficiency of the resonator element **6** is improved. Therefore, it is possible to stabilize a temperature of the resonator element **6** and save power. However, an atmosphere of the accommodating space **Q2** is not particularly limited.

(31) The outer base **21** includes a plurality of internal terminals **241** disposed at the bottom surface of the first upper recessed portion **211a**, a plurality of internal terminals **242** disposed at the bottom surface of the second upper recessed portion **211b**, a plurality of internal terminals **243** disposed at a bottom surface of the lower recessed portion **212**, and a plurality of external terminals **244** disposed at the lower surface of the outer base **21**. These terminals are electrically coupled via an internal wiring (not illustrated) formed in the outer base **21** so as to implement a desired circuit. The internal terminals **241** are electrically coupled to the circuit element **4** via bonding wires **BW1**, and the internal terminals **242** are electrically coupled to the inner package **3** via bonding wires

BW2.

(32) The outer package **2** is described above. The outer base **21** corresponds to a first container of the first oscillator **1A**, a second container of the second oscillator **1B**, a fifth container of the third oscillator **1C**, and a seventh container of the fourth oscillator **1D**. That is, it means that outer bases **21** of the same type are used as the first container, the second container, the fifth container, and the seventh container. The outer bases **21** of the same type mean that the outer bases **21** have the same outer shape, the same shape and dimension of each portion such as the recessed portion, the electrode, and the wiring, and the same material (component material) of each portion. Further, the outer bases **21** of the same type may mean outer packages having the same model number (identification code).

(33) The upper recessed portion **211** corresponds to a first recessed portion of the first oscillator **1A**, a third recessed portion of the second oscillator **1B**, a fifth recessed portion of the third oscillator **1C**, and a seventh recessed portion of the fourth oscillator **1D**. The lower recessed portion **212** corresponds to a second recessed portion of the first oscillator **1A**, a fourth recessed portion of the second oscillator **1B**, a sixth recessed portion of the third oscillator **1C**, and an eighth recessed portion of the fourth oscillator **1D**.

(34) Inner Package **3**

(35) As illustrated in FIG. **3**, the inner package **3** includes an inner base **31** and an inner lid **32**. The inner base **31** has a box shape, and includes a recessed portion **311** that is open in a lower surface thereof. The inner lid **32** has a plate shape, and is bonded to the lower surface of the inner base **31** via a sealing member **33** such as a seal ring or low-melting glass so as to close an opening of the recessed portion **311**. Accordingly, the recessed portion **311** is hermetically sealed, and an accommodating space **Q3** is formed in the inner package **3**. The resonator element **6**, the temperature control element **7**, and the circuit element **8** are accommodated in the accommodating space **Q3**.

(36) Component materials of the inner base **31** and the inner lid **32** are not particularly limited. For example, the inner base **31** may be made of various ceramic materials containing alumina, titania or the like, and the inner lid **32** may be made of various metal materials such as Kovar.

(37) The accommodating space **Q3** will be described in detail, the recessed portion **311** includes a first recessed portion **311a** that is open in the lower surface, a second recessed portion **311b** that is open in a bottom surface of the first recessed portion **311a** and has an opening smaller than that of the first recessed portion **311a**, and a third recessed portion **311c** that is open in a bottom surface of the second recessed portion **311b** and has an opening smaller than that of the second recessed portion **311b**. The resonator element **6** is disposed at the bottom surface of the first recessed portion **311a**, and the temperature control element **7** and the circuit element **8** are disposed side by side at a bottom surface of the third recessed portion **311c**.

(38) The accommodating space **Q3** is airtight and is in a depressurized state, preferably, in a state close to vacuum. Accordingly, viscous resistance of the accommodating space **Q3** is reduced, and vibration characteristics of the resonator element **6** are improved. However, an atmosphere of the accommodating space **Q3** is not particularly limited.

(39) The inner base **31** includes a plurality of internal terminals **341** disposed at the bottom surface of the first recessed portion **311a**, a plurality of internal terminals **342** and **343** disposed at the bottom surface of the second recessed portion **311b**, and a plurality of external terminals **344** disposed at an upper surface of the inner base **31**. These terminals are electrically coupled via an internal wiring (not illustrated) formed in the inner base **31** so as to implement a desired circuit.

(40) The internal terminals **341** are electrically coupled to the resonator element **6** via a conductive bonding member **B2** and a bonding wire **BW3**. The internal terminals **342** are electrically coupled to the temperature control element **7** via a bonding wire **BW4**. The internal terminals **343** are electrically coupled to the circuit element **8** via bonding wires **BW5**. The external terminals **344** are electrically coupled to the outer package **2** via the bonding wires **BW2**.

(41) The inner package **3** as described above is fixed to the bottom surface of the third upper recessed portion **211c** via a bonding member **B3** at the inner lid **32**.

(42) Resonator Element **6**

(43) The resonator element **6** is an SC-cut quartz crystal resonator element. Since the SC-cut quartz crystal resonator element has a third-order frequency-temperature characteristic, the resonator element **6** has excellent frequency stability. In particular, since the SC-cut quartz crystal resonator element has an inflection point around 95° C., the SC-cut quartz crystal resonator element is suitable for an oven-controlled crystal oscillator (OCXO) that is used to heat the resonator element **6**. Therefore, the oscillator **1** with high accuracy can be manufactured at low cost using the common package **100**. The SC cut refers to a cut angle in cutting from a plane obtained by rotating a plane orthogonal to a Y-axis (mechanical axis), which is a crystal axis of quartz crystal, by about 33° to 35° around an X-axis (electric axis) and further rotating the rotated plane by about 22° to 24° around a Z-axis (optical axis).

(44) As illustrated in FIG. **4**, the resonator element **6** includes a quartz-crystal substrate **61** cut out by SC cut and having a rectangular shape in a plan view, an excitation electrode **621** disposed at a central portion of an upper surface, a pad electrode **622** extracted from the excitation electrode **621** and disposed at an edge portion of the upper surface, an excitation electrode **631** disposed to face the excitation electrode **621** at a central portion of a lower surface, and a pad electrode **632** extracted from the excitation electrode **631** and disposed at an edge portion of the lower surface.

(45) Although the resonator element **6** is described above, a configuration thereof is not particularly limited. For example, a planar shape of the quartz crystal substrate **61** is not limited to a rectangular shape, and may be a circular shape, an elliptical shape, a semicircular shape, or another polygonal shape. In addition, bevel processing of grinding an outer edge portion of the quartz-crystal substrate **61** or convex processing of forming the upper surface and the lower surface of the quartz-crystal substrate **61** into convex curved surfaces may be performed.

(46) The resonator element **6** may be an AT-cut quartz crystal resonator element. Since the AT-cut quartz crystal resonator element has a third-order frequency-temperature characteristic, the resonator element **6** has excellent frequency stability as in the case of the SC-cut quartz crystal resonator element. In particular, since the AT-cut quartz crystal resonator element has a higher demand and a larger circulation volume than the SC-cut quartz crystal resonator element, cost reduction can be achieved.

(47) Instead of the SC-cut quartz crystal resonator element and the AT-cut quartz crystal resonator element, a BT-cut quartz crystal resonator element, a tuning-fork type quartz crystal resonator element, a surface acoustic wave resonator, another piezoelectric resonator element, a MEMS resonance element, or the like may be used as the resonator element **6**.

(48) An end portion of the resonator element **6** is fixed at the bottom surface of the first recessed portion **311a** via the bonding member **B2**. The pad electrodes **622** and **632** are electrically coupled to the internal terminals **341** via the bonding member **B2** and the bonding wire **BW3**. However, a fixing method and an electrical coupling method of the resonator element **6** are not particularly limited.

(49) Temperature Control Element **7**

(50) As illustrated in FIG. **3**, the temperature control element **7** includes a temperature sensor **71** and a heating circuit **72**. The temperature sensor **71** functions as a temperature detection unit configured to detect an ambient temperature, particularly a temperature of the resonator element **6**, and the heating circuit **72** functions as a heating unit configured to heat the resonator element **6**. The temperature control element **7** is disposed at the bottom surface of the third recessed portion **311c**, and is electrically coupled to the plurality of internal terminals **342** via the bonding wires **BW4**. In the embodiment, since the resonator element **6** and the temperature control element **7** are accommodated in the same space, a difference between a detection result of the temperature sensor **71** and an actual temperature of the resonator element **6** is reduced.

(51) As described above, by providing the temperature control element **7**, the common package **100** capable of outputting a high-accuracy oscillation signal with little frequency fluctuation with respect to a change in environmental temperature is obtained. By using such a common package **100**, it is possible to efficiently manufacture the oscillators **1A**, **1B**, **1C**, and **1D** at low cost.

(52) Circuit Element **8**

(53) As illustrated in FIG. **3**, the circuit element **8** includes an oscillation circuit **81** configured to oscillate the resonator element **6**. The oscillation circuit **81** is a circuit for amplifying a signal output from the resonator element **6** and feeding back the amplified signal to the resonator element **6**, thereby causing the resonator element **6** to oscillate to generate an oscillation signal. The circuit element **8** further includes a temperature compensation circuit **82**. The temperature compensation circuit **82** performs temperature compensation based on temperature information output from the temperature sensor **71** and temperature compensation data such that the frequency fluctuation of the oscillation signal is smaller than the frequency-temperature characteristic of the resonator element **6**, preferably such that the oscillation signal is constant within a predetermined temperature range. In the circuit element **8**, whether to use the temperature compensation circuit **82** can be set according to a user's desire.

(54) In the embodiment, a mode in which the temperature compensation is performed using the temperature sensor **71** of the temperature control element **7** is described. Alternatively, a temperature sensor may be built in the circuit element **8**, and the temperature compensation may be performed using the temperature sensor. Specifically, the temperature compensation may be performed based on temperature information output from the temperature sensor built in the circuit element **8** and the temperature compensation data such that the frequency fluctuation of the oscillation signal is smaller than the frequency-temperature characteristic of the resonator element **6**, preferably such that the oscillation signal is constant within a predetermined temperature range.

(55) Circuit Element **4**

(56) As illustrated in FIG. **1**, the circuit element **4** includes a temperature control circuit **41** configured to control driving of the temperature control element **7**, a part of a PLL circuit **42**, and an output buffer circuit **43**.

(57) The temperature control circuit **41** is a circuit for controlling an amount of current flowing through a resistor of the heating circuit **72** based on an output signal of the temperature sensor **71** to keep the resonator element **6** at a constant temperature. For example, when a current temperature determined based on the output signal of the temperature sensor **71** is lower than a set reference temperature, the temperature control circuit **41** performs control such that a desired current flows through the resistor of the heating circuit **72**. When the current temperature is higher than the reference temperature, the temperature control circuit **41** performs control such that the current does not flow through the resistor of the heating circuit **72**. In addition, for example, the temperature control circuit **41** performs control to increase or decrease the amount of current flowing through the resistor of the heating circuit **72** according to a difference between the current temperature and the reference temperature.

(58) As illustrated in FIG. **5**, the circuit element **4** includes, as a part of the PLL circuit **42**, a first phase comparator **421**, a first low-pass filter **422**, a voltage-controlled oscillator **423**, a first divider **424**, a second phase comparator **425**, a second low-pass filter **426**, an oscillation circuit **427**, and a second divider **428**. In order to enhance versatility, the circuit element **4** further includes a first switch circuit **SW1** disposed between the voltage-controlled oscillator **423** and the second phase comparator **425** and a second switch circuit **SW2** disposed between the second low-pass filter **426** and the oscillation circuit **427**.

(59) In such a circuit element **4**, a first route **R1** illustrated in FIG. **6**, a second route **R2** illustrated in FIG. **7**, or a third route **R3** illustrated in FIG. **8** can be set by switching between routes using the first and second switch circuits **SW1** and **SW2**. Accordingly, the circuit element **4** having high versatility is obtained. The route may be set, for example, by writing, in the circuit element **4**, a

program corresponding to a route to be set, or by writing programs corresponding to all routes in advance and selecting a predetermined program from the written programs.

(60) The common package **100** is described above. Next, the first oscillator **1A**, the second oscillator **1B**, the third oscillator **1C**, and the fourth oscillator **1D** manufactured using the common package **100** will be described in this order.

(61) First Oscillator **1A**

(62) The first oscillator **1A** is an oven-controlled crystal oscillator (OCXO). Therefore, the temperature compensation circuit **82** provided in the circuit element **8** is not used. The circuit element **4** is set to the first route **R1**. As illustrated in FIG. **9**, the first oscillator **1A** has a configuration in which a voltage-controlled crystal oscillator **5** is disposed in the common package **100**. The voltage-controlled crystal oscillator **5** is disposed in the lower recessed portion **212**, and is electrically coupled to the internal terminals **243** via a conductive bonding member **B1**.

(63) As described above, since the lower recessed portion **212** is formed in the lower surface of the outer package **2** and the voltage-controlled crystal oscillator **5** is mounted to the lower recessed portion **212**, mounting of the voltage-controlled crystal oscillator **5** is facilitated. In addition, according to such a configuration, a package in which the upper recessed portion **211** is sealed with the outer lid **22** can be set as the common package **100**. Therefore, the number of components provided in the common package **100** is increased, and the number of individual steps for each of the oscillators **1A**, **1B**, **1C**, and **1D** is reduced accordingly, and the oscillators **1A**, **1B**, **1C**, and **1D** can be manufactured efficiently.

(64) As illustrated in FIG. **6**, the voltage-controlled crystal oscillator **5** is an oscillator provided in the PLL circuit **42**. As illustrated in FIG. **10**, the voltage-controlled crystal oscillator **5** includes a package **51**, and a resonator element **55** and a circuit element **59** accommodated in the package **51**. In this way, by accommodating the resonator element **55** and the circuit element **59** in the package **51**, the resonator element **55** and the circuit element **59** can be collectively mounted to the outer base **21**. The resonator element **55** and the circuit element **59** can also be protected by the package **51**.

(65) The package **51** corresponds to a third container of the first oscillator **1A** and a fourth container of the second oscillator **1B**. The resonator element **55** corresponds to a second resonator element of the first oscillator **1A** and a fourth resonator element of the second oscillator **1B**. The circuit element **59** corresponds to a third circuit element of the first oscillator **1A** and a fourth circuit element of the second oscillator **1B**.

(66) The package **51** includes a base **52** and a lid **53**. The base **52** has a box shape and includes a recessed portion **521** that is open in a lower surface thereof. The lid **53** has a plate shape, and is bonded to the lower surface of the base **52** via a sealing member **54** such as a seal ring or low-melting glass so as to close an opening of the recessed portion **521**. Accordingly, the recessed portion **521** is hermetically sealed, and an accommodating space **Q5** is formed in the package **51**. The resonator element **55** and the circuit element **59** are accommodated in the accommodating space **Q5**.

(67) Component materials of the base **52** and the lid **53** are not particularly limited. For example, the base **52** may be made of various ceramic materials containing alumina, titania or the like, and the lid **53** may be made of various metal materials such as Kovar.

(68) The accommodating space **Q5** will be described in detail, the recessed portion **521** includes a first recessed portion **521a** that is open in the lower surface, a second recessed portion **521b** that is open in a bottom surface of the first recessed portion **521a** and has an opening smaller than that of the first recessed portion **521a**, and a third recessed portion **521c** that is open in a bottom surface of the second recessed portion **521b** and has an opening smaller than that of the second recessed portion **521b**. The resonator element **55** is disposed at the bottom surface of the first recessed portion **521a**, and the circuit element **59** is disposed at a bottom surface of the third recessed portion **521c**. The accommodating space **Q5** is airtight and is in a depressurized state, preferably in

a state close to vacuum. Accordingly, viscous resistance of the accommodating space Q5 is reduced, and vibration characteristics of the resonator element 55 are improved. However, an atmosphere of the accommodating space Q5 is not particularly limited.

(69) The base 52 includes a plurality of internal terminals 561 disposed at the bottom surface of the first recessed portion 521a, a plurality of internal terminals 562 disposed at the bottom surface of the second recessed portion 521b, and a plurality of external terminals 564 disposed at an upper surface of the base 52. These terminals are electrically coupled via an internal wiring (not illustrated) formed in the base 52 so as to implement a desired circuit. The internal terminals 561 are electrically coupled to the resonator element 55 via a conductive bonding member B4. The internal terminals 562 are electrically coupled to the circuit element 59 via bonding wires BW6. The external terminals 564 are electrically coupled to the internal terminals 243 via the bonding member B1.

(70) The resonator element 55 is an AT-cut quartz crystal resonator element. Since the AT-cut quartz crystal resonator element has a third-order frequency-temperature characteristic, the resonator element 55 has excellent frequency stability. Therefore, the first oscillator LA with high accuracy is obtained. The AT cut refers to a cut angle in cutting from a plane obtained by rotating a plane orthogonal to a Y-axis (mechanical axis), which is a crystal axis of quartz crystal, by about 33° to 36° around an X-axis (electric axis).

(71) As illustrated in FIG. 11, the resonator element 55 includes an oblong quartz-crystal substrate 551 cut out by AT-cut, excitation electrodes 553a and 554a arranged to face each other at upper and lower surfaces of the quartz-crystal substrate 551, pad electrodes 553b and 554b arranged at the upper surface of the quartz-crystal substrate 551, and extraction electrodes 553c and 554c coupling the excitation electrodes 553a and 554a and the pad electrodes 553b and 554b.

(72) Although the resonator element 55 is described above, a configuration thereof is not particularly limited. For example, a planar shape of the quartz-crystal substrate 551 is not limited to an oblong shape, and may be a circular shape, a rectangular shape other than an oblong shape, or another polygonal shape. In addition, bevel processing of grinding an outer edge portion of the quartz-crystal substrate 551 or convex processing of forming the upper surface and the lower surface of the quartz-crystal substrate 551 into convex curved surfaces may be performed. Instead of the AT-cut quartz crystal resonator element, an SC-cut quartz crystal resonator element, a BT-cut quartz crystal resonator element, a tuning-fork type quartz crystal resonator element, a surface acoustic wave resonator element, another piezoelectric resonator element, a MEMS resonance element, or the like may be used as the resonator element 55.

(73) An end portion of the resonator element 55 is fixed at the bottom surface of the first recessed portion 521a via the bonding member B4. The pad electrodes 553b and 554b are electrically coupled to the internal terminals 561 via the bonding member B4. However, a fixing method and an electrical coupling method of the resonator element 55 are not particularly limited.

(74) The circuit element 59 includes an oscillation circuit 591 configured to oscillate the resonator element 55. The oscillation circuit 591 is electrically coupled to the resonator element 55, amplifies an output signal of the resonator element 55, and feeds back the amplified signal to the resonator element 55 to oscillate the resonator element 55.

(75) The configuration of the first oscillator 1A is described above. In the first oscillator 1A, the PLL circuit 42 operates as follows. Since the first oscillator 1A includes the voltage-controlled crystal oscillator 5, the oscillation circuit 427 in the PLL circuit 42 is not necessary. Therefore, as described above, the first route R1 illustrated in FIG. 6 is set.

(76) The first phase comparator 421 detects a phase difference between a first oscillation signal V1, which is a reference frequency signal output from the oscillation circuit 81, and a frequency signal output from the first divider 424, and outputs the detected phase difference to the first low-pass filter 422. The first low-pass filter 422 removes the high-frequency component from the output signal from the first phase comparator 421, converts the signal into a voltage, and outputs the

voltage as a DC signal for controlling the voltage-controlled oscillator **423**. The first divider **424** outputs a frequency signal, which is obtained by dividing a frequency signal output from the voltage-controlled oscillator **423**, to the first phase comparator **421**.

(77) For example, the first divider **424** can set a decimal division ratio by switching an integer division ratio to a decimal division ratio on average. Accordingly, a PLL circuit portion at an upstream stage including the first phase comparator **421**, the first low-pass filter **422**, the voltage-controlled oscillator **423**, and the first divider **424** functions as a decimal division PLL circuit (fractional PLL circuit). As a result, in the decimal division PLL circuit, a signal of any frequency can be output.

(78) The second phase comparator **425** detects a phase difference between a frequency signal output from the voltage-controlled oscillator **423** and a frequency signal output from the second divider **428**, and outputs the detected phase difference to the second low-pass filter **426**. The second low-pass filter **426** removes the high-frequency component from an output signal from the second phase comparator **425**, converts the signal into a voltage, and outputs a DC signal for controlling the voltage-controlled crystal oscillator **5**. The second divider **428** outputs a frequency signal, which is obtained by dividing a frequency signal output from the voltage-controlled crystal oscillator **5**, to the second phase comparator **425**.

(79) The second divider **428** is, for example, an integer divider configured to divide an input signal by an integer. Accordingly, a PLL circuit portion at a downstream stage including the second phase comparator **425**, the second low-pass filter **426**, the voltage-controlled crystal oscillator **5**, and the second divider **428** functions as an integer division PLL circuit (integer PLL circuit). In the integer division PLL circuit, phase noise is relatively little, and a circuit configuration can be relatively simple.

(80) The voltage-controlled crystal oscillator **5** outputs a third oscillation signal **V3**, which is a frequency signal corresponding to the voltage of the DC signal, to the output buffer circuit **43**. That is, the PLL circuit **42** oscillates the resonator element **55** to generate the third oscillation signal **V3** whose frequency is controlled based on the first oscillation signal **V1**. Accordingly, the first oscillator **1A** with high accuracy is obtained.

(81) Second Oscillator **1B**

(82) The second oscillator **1B** illustrated in FIG. **12** has the same configuration as the first oscillator **1A** described above except that the configuration of the voltage-controlled crystal oscillator **5** is different. In the second oscillator **1B**, a frequency (oscillation frequency) of the resonator element **55** is different from that in the first oscillator **1A**. For example, while the oscillation frequency of the resonator element **55** is 20 MHz in the first oscillator **1A**, the oscillation frequency of the resonator element **55** is 40 MHz in the second oscillator **1B**.

(83) The configuration of the second oscillator **1B** is described above. In the second oscillator **1B**, the PLL circuit **42** operates as follows.

(84) As illustrated in FIG. **6**, the first phase comparator **421** detects a phase difference between a second oscillation signal **V2**, which is a reference frequency signal output from the oscillation circuit **81**, and a frequency signal output from the first divider **424**, and outputs the detected phase difference to the first low-pass filter **422**. The first low-pass filter **422** removes the high-frequency component from an output signal from the first phase comparator **421**, converts the signal into a voltage, and outputs the voltage as a DC signal for controlling the voltage-controlled oscillator **423**. The first divider **424** outputs a frequency signal, which is obtained by dividing a frequency signal output from the voltage-controlled oscillator **423**, to the first phase comparator **421**.

(85) The second phase comparator **425** detects a phase difference between a frequency signal output from the voltage-controlled oscillator **423** and a frequency signal output from the second divider **428**, and outputs the detected phase difference to the second low-pass filter **426**. The second low-pass filter **426** removes the high-frequency component from the output signal from the second phase comparator **425**, converts the signal into a voltage, and outputs a DC signal for controlling

the voltage-controlled crystal oscillator **5**. The second divider **428** outputs a frequency signal, which is obtained by dividing a frequency signal output from the voltage-controlled crystal oscillator **5**, to the second phase comparator **425**.

(86) The voltage-controlled crystal oscillator **5** outputs a fourth oscillation signal **V4**, which is a frequency signal corresponding to the voltage of the DC signal, to the output buffer circuit **43**. That is, the PLL circuit **42** oscillates the resonator element **55** to generate the fourth oscillation signal **V4** whose frequency is controlled based on the second oscillation signal **V2**. Accordingly, the second oscillator **1B** with high accuracy is obtained.

(87) Third Oscillator **1C**

(88) The third oscillator **1C** illustrated in FIG. **13** is the same as the above-described first oscillator **1A** except that the voltage-controlled crystal oscillator **5** is omitted and the PLL circuit **42** is set to the third route **R3**. That is, the third oscillator **1C** includes only the common package **100**.

Accordingly, the third oscillator **1C** can be manufactured at low cost. In the third oscillator **1C**, the PLL circuit **42** operates as follows.

(89) As illustrated in FIG. **8**, the first phase comparator **421** detects a phase difference between a fifth oscillation signal **V5**, which is a reference frequency signal output from the oscillation circuit **81**, and a frequency signal output from the first divider **424**, and outputs the detected phase difference to the first low-pass filter **422**. The first low-pass filter **422** removes a high-frequency component from the output signal from the first phase comparator **421**, converts the signal into a voltage, and outputs the voltage as a DC signal for controlling the voltage-controlled oscillator **423**. The first divider **424** outputs a frequency signal, which is obtained by dividing a frequency signal output from the voltage-controlled oscillator **423**, to the first phase comparator **421**. The voltage-controlled oscillator **423** outputs a sixth oscillation signal **V6**, which is a frequency signal corresponding to the voltage of the DC signal, to the output buffer circuit **43**.

(90) Fourth Oscillator **1D**

(91) The fourth oscillator **1D** illustrated in FIG. **14** is the same as the above-described first oscillator **1A** except that a quartz crystal resonator **50** is used instead of the voltage-controlled crystal oscillator **5** and the circuit element **4** is set to the second route **R2**. The quartz crystal resonator **50** has a configuration obtained by omitting the circuit element **59** from the voltage-controlled crystal oscillator **5**. Instead, as illustrated in FIG. **7**, the oscillation circuit **427** of the circuit element **4** is electrically coupled to the resonator element **55**. In the fourth oscillator **1D**, the PLL circuit **42** operates as follows.

(92) As illustrated in FIG. **7**, the first phase comparator **421** detects a phase difference between a seventh oscillation signal **V7**, which is a reference frequency signal output from the oscillation circuit **81**, and a frequency signal output from the first divider **424**, and outputs the detected phase difference to the first low-pass filter **422**. The first low-pass filter **422** removes a high-frequency component from the output signal from the first phase comparator **421**, converts the signal into a voltage, and outputs the voltage as a DC signal for controlling the voltage-controlled oscillator **423**. The first divider **424** outputs a frequency signal, which is obtained by dividing a frequency signal output from the voltage-controlled oscillator **423**, to the first phase comparator **421**.

(93) The second phase comparator **425** detects a phase difference between a frequency signal output from the voltage-controlled oscillator **423** and a frequency signal output from the second divider **428**, and outputs the detected phase difference to the second low-pass filter **426**. The second low-pass filter **426** removes a high-frequency component from the output signal from the second phase comparator **425**, converts the signal into a voltage, and outputs a DC signal for controlling the oscillation circuit **427**. The second divider **428** outputs a frequency signal, which is obtained by dividing a frequency signal output from the oscillation circuit **427**, to the second phase comparator **425**.

(94) The oscillation circuit **427** outputs an eighth oscillation signal **V8**, which is a frequency signal corresponding to the voltage of the DC signal, to the output buffer circuit **43**. That is, the PLL

circuit **42** oscillates the resonator element **55** to generate the eighth oscillation signal **V8** whose frequency is controlled based on the seventh oscillation signal **V7**. Accordingly, the fourth oscillator **1D** with high accuracy is obtained.

(95) The oscillators **1A**, **1B**, **1C**, and **1D** are described above. Next, a method for manufacturing the plurality of oscillators **1** (**1A**, **1B**, **1C**, and **1D**) will be described. As illustrated in FIG. **15**, a method for manufacturing a plurality of oscillators **1** includes: a step **S1** of preparing a plurality of common packages **100**; a step **S2** of manufacturing the first oscillator **1A** by mounting the voltage-controlled crystal oscillator **5** having a predetermined oscillation frequency to the lower recessed portion **212** of the common package **100** and setting the PLL circuit **42** to the first route **R1**; a step **S3** of manufacturing the second oscillator **1B** by mounting the voltage-controlled crystal oscillator **5**, which has an oscillation frequency different from that of the above voltage-controlled crystal oscillator **5** mounted to the first oscillator **1A**, to the lower recessed portion **212** of the common package **100** and setting the PLL circuit **42** to the first route **R1**; a step **S4** of manufacturing the third oscillator **1C** by setting the PLL circuit **42** to the third route **R3**; and a step **S5** of manufacturing the fourth oscillator **1D** by mounting the quartz crystal resonator **50** to the lower recessed portion **212** of the common package **100** and setting the PLL circuit **42** to the second route **R2**.

(96) In manufacturing of the four types of oscillators **1A**, **1B**, **1C**, and **1D**, after the step **S1** is performed, the step **S2** is performed when a manufacturing target is the first oscillator **1A**, the step **S3** is performed when the manufacturing target is the second oscillator **1B**, the step **S4** is performed when the manufacturing target is the third oscillator **1C**, and the step **S5** is performed when the manufacturing target is the fourth oscillator **1D**.

(97) The step **S1** includes: a step **S11** of preparing the outer base **21**; a step **S12** of mounting the inner package **3**, in which the resonator element **6**, the temperature control element **7**, and the circuit element **8** are accommodated, to the upper recessed portion **211** of the outer base **21**; a step **S13** of mounting the circuit element **4** to the upper recessed portion **211** of the outer base **21**; and a step **S14** of bonding the outer lid **22** to the outer base **21** to obtain the common package **100**.

(98) Here, the manufacturing method, in other words, the method for manufacturing the oscillators **1**, as illustrated in FIG. **16**, includes manufacturing the first oscillator **1A**, the manufacturing the first oscillator **1A** including a first-oscillator first step **Sa1** of mounting the resonator element **6**, the temperature control element **7**, and the circuit element **8** to the upper recessed portion **211** of the outer base **21**, a first-oscillator second step **Sa2** of mounting the voltage-controlled crystal oscillator **5** to the lower recessed portion **212** of the outer base **21**, and a first-oscillator third step **Sa3** of setting the PLL circuit **42**; manufacturing the second oscillator **1B**, the manufacturing the second oscillator **1B** including a second-oscillator first step **Sb1** of mounting the resonator element **6**, the temperature control element **7**, and the circuit element **8** to the upper recessed portion **211** of the outer base **21**, a second-oscillator second step **Sb2** of mounting the voltage-controlled crystal oscillator **5** to the lower recessed portion **212** of the outer base **21**, and a second-oscillator third step **Sb3** of setting the PLL circuit **42**; manufacturing the third oscillator **1C**, the manufacturing the third oscillator **1C** including a third-oscillator first step **Sc1** of mounting the resonator element **6**, the temperature control element **7**, and the circuit element **8** to the upper recessed portion **211** of the outer base **21**, and a third-oscillator second step **Sc2** of setting the PLL circuit **42**; and manufacturing the fourth oscillator **1D**, the manufacturing the fourth oscillator **1D** including a fourth-oscillator first step **Sd1** of mounting the resonator element **6**, the temperature control element **7**, and the circuit element **8** to the upper recessed portion **211** of the outer base **21**, a fourth-oscillator second step **Sd2** of mounting the quartz crystal resonator **50** to the lower recessed portion **212** of the outer base **21**, and a fourth-oscillator third step **Sd3** of setting the PLL circuit **42**.

(99) According to the method for manufacturing the oscillators **1**, since the common package **100** is used, it is possible to efficiently manufacture a plurality of oscillators **1** (**1A**, **1B**, **1C**, and **1D**) of different types at low cost.

(100) Although the four types of oscillators **1A**, **1B**, **1C**, and **1D** are manufactured using the common package **100** in the embodiment, the present disclosure is not limited thereto, and at least two oscillators having different output frequencies may be manufactured using the common package **100**. In the example of the embodiment, the first oscillator **1A** and the second oscillator **1B** may be manufactured. In this case, since only the first route **R1** is used in the PLL circuit **42**, the oscillation circuit **427**, the first switch circuit **SW1**, and the second switch circuit **SW2** can be omitted from the PLL circuit **42**. Further, the third oscillator **1C** and a fifth oscillator, in which the frequency of the resonator element **55** is different from that in the third oscillator **1C**, may be manufactured. In this case, since only the second route **R2** is used in the PLL circuit **42**, the first switch circuit **SW1** and the second switch circuit **SW2** can be omitted from the PLL circuit **42**.

(101) Although the oscillator **1** that is an oven-controlled crystal oscillator (OCXO) is manufactured using the common package **100** in the embodiment, the oscillator **1** is not limited to the oven-controlled crystal oscillator (OCXO). For example, the oscillator **1** that is a simple packaged crystal oscillator (SPXO) may be manufactured by setting the temperature control element **7** and the temperature compensation circuit **82** not to be used together, or the oscillator **1** that is a temperature-compensated crystal oscillator (TCXO) may be manufactured by setting the temperature control element **7** not to be used and the temperature compensation circuit **82** to be used. When manufacturing only the oscillator **1** that is an oven-controlled crystal oscillator, the temperature compensation circuit **82** may be omitted from the circuit element **8**.

(102) In the embodiment, the resonator element **55** and the circuit element **59** are mounted to the outer base **21** in a state of being accommodated in the package **51** in the embodiment. Alternatively, the package **51** may be omitted, and the resonator element **55** and the circuit element **59** may be mounted to the outer base **21** separately. In this case, the opening of the lower recessed portion **212** is preferably covered with a lid to form an accommodating space. According to such a configuration, a reduction in size of common package **100** can be achieved.

(103) The method for manufacturing the oscillators **1** is described above. As described above, the method for manufacturing the oscillators **1** is a method for manufacturing the oscillators **1** of a plurality of types including the first oscillator **1A** and the second oscillator **1B**, the method including: manufacturing the first oscillator **1A**, the manufacturing the first oscillator **1A** including the first-oscillator first step **Sa1** of mounting the resonator element **6** and the circuit element **8** to the outer base **21** serving as a first container, the resonator element **6** being a first resonator element and the circuit element **8** being a first circuit element configured to oscillate the resonator element **6** to generate the first oscillation signal **V1**, and the first-oscillator second step **Sa2** of mounting, to the outer base **21**, the resonator element **55** that is a second resonator element whose oscillation frequency is controlled based on the first oscillation signal **V1**; and manufacturing the second oscillator **1B**, the manufacturing the second oscillator **1B** including the second-oscillator first step **Sb1** of mounting the resonator element **6** and the circuit element **8** to the outer base **21** that is a second container of the same type as the outer base **21** of the first oscillator **1A**, the resonator element **6** being a third resonator element of the same type as the resonator element **6** of the first oscillator **1A** and the circuit element **8** being a second circuit element of the same type as the circuit element **8** of the first oscillator **1A** and being configured to oscillate the resonator element **6** to generate the second oscillation signal **V2**, and the second-oscillator second step **Sb2** of mounting the resonator element **55** to the outer base **21**, the resonator element **55** being a fourth resonator element whose oscillation frequency is controlled based on the second oscillation signal **V2** and whose frequency is different from that of the resonator element **55**. According to such a manufacturing method, since the first-oscillator first step **Sa1** and the second-oscillator first step **Sb1** are common, the first oscillator **1A** and the second oscillator **1B** having different output frequencies can be efficiently manufactured at low cost.

(104) Further, as described above, the outer base **21** serving as the first container includes the upper recessed portion **211** that is a first recessed portion opening in one main surface, and the lower

recessed portion **212** that is a second recessed portion opening in the other main surface. In the first-oscillator first step Sa1, the resonator element **6** and the circuit element **8** are mounted to the upper recessed portion **211**, and in the first-oscillator second step Sa2, the resonator element **55** is mounted to the lower recessed portion **212**. The outer base **21** serving as the second container includes the upper recessed portion **211** that is a third recessed portion opening in one main surface, and the lower recessed portion **212** that is a fourth recessed portion opening in the other main surface. In the second-oscillator first step Sb1, the resonator element **6** and the circuit element **8** are mounted to the upper recessed portion **211**, and in the second-oscillator second step Sb2, the resonator element **55** is mounted to the lower recessed portion **212**. According to such a method, it is possible to increase the number of steps that can be performed in common between the first-oscillator first step Sa1 and second-oscillator first step Sb1. Therefore, the number of steps in the first-oscillator second step Sa2 and the second-oscillator second step Sb2, which are individual steps, can be reduced. Accordingly, the first oscillator **1A** and the second oscillator **1B** having different output frequencies can be efficiently manufactured at low cost.

(105) Further, as described above, in the first-oscillator first step Sa1, the temperature control element **7** that is a first heater circuit configured to heat the resonator element **6** is further mounted to the outer base **21**, and in the second-oscillator first step Sb1, the temperature control element **7** that is a second heater circuit of the same type as the temperature control element **7** configured to heat the resonator element **6** is further mounted to the outer base **21**. As described above, by providing the temperature control element **7**, it is possible to manufacture the first oscillator **1A** and the second oscillator **1B** capable of outputting a high-accuracy oscillation signal with little frequency fluctuation with respect to a change in environmental temperature.

(106) As described above, each of the resonator element **6** of the first oscillator **1A** and the resonator element **6** of the second oscillator **1B** is an SC-cut quartz crystal resonator element. Since the SC-cut quartz crystal resonator element has a third-order frequency-temperature characteristic, the resonator element **6** has excellent frequency stability. In particular, since the SC-cut quartz crystal resonator element has an inflection point around 95° C., the SC-cut quartz crystal resonator element is suitable for an oven-controlled crystal oscillator (OCXO) that is used to heat the resonator element **6**.

(107) As described above, each of the resonator element **6** of the first oscillator **1A** and the resonator element **6** of the second oscillator **1B** is an AT-cut quartz crystal resonator element. Since the AT-cut crystal resonator element has a third-order frequency-temperature characteristic, the resonator element **6** has excellent frequency stability. In particular, since the AT-cut quartz crystal resonator element has a higher demand and a larger circulation volume than the SC-cut quartz crystal resonator element, cost reduction can be achieved.

(108) As described above, each of the resonator element **55** of the first oscillator **1A** and the resonator element **55** of the second oscillator **1B** is an AT-cut quartz crystal resonator element. Since the AT-cut crystal resonator element has a third-order frequency-temperature characteristic, the resonator element **55** has excellent frequency stability.

(109) Further, as described above, in the first-oscillator second step Sa2, the circuit element **59** that is a third circuit element configured to oscillate the resonator element **55** to generate the third oscillation signal **V3** whose frequency is controlled based on the first oscillation signal **V1** is further mounted to the outer base **21**, and in the second-oscillator second step Sb2, the circuit element **59** that is a fourth circuit element configured to oscillate the resonator element **55** to generate the fourth oscillation signal **V4** whose frequency is controlled based on the second oscillation signal **V2** is further mounted to the outer base **21**. Accordingly, the third oscillation signal **V3** and the fourth oscillation signal **V4** with little phase noise can be generated.

(110) As described above, in the first-oscillator second step Sa2, the resonator element **55** and the circuit element **59** are mounted to the outer base **21** in a state of being accommodated in the package **51** serving as a third container, and in the second-oscillator second step Sb2, the resonator

element **55** and the circuit element **59** are mounted to the outer base **21** in a state of being accommodated in the package **51** serving as a fourth container. As described above, since the resonator element **55** and the circuit element **59** can be collectively mounted by using the package **51**, handleability is improved. Therefore, the first-oscillator second step Sa2 and the second-oscillator second step Sb2 can be smoothly performed. The resonator element **55** and the circuit element **59** can also be protected by the package **51**.

(111) As described above, the method for manufacturing the oscillator **1** includes manufacturing the third oscillator **1C**, the manufacturing the third oscillator **1C** including the third-oscillator first step Sc1 of mounting the resonator element **6** and the circuit element **8** to the outer base **21** that is a fifth container of the same type as the outer base **21** of the first oscillator **1A**, the resonator element **6** being a fifth resonator element of the same type as the resonator element **6** of the first oscillator **1A**, and the circuit element **8** being a fifth circuit element of the same type as the circuit element **8** of the first oscillator **1A** and being configured to oscillate the resonator element **6** to generate the fifth oscillation signal V5. According to such a manufacturing method, since the first-oscillator first step Sa1, the second-oscillator first step Sb1, and the third-oscillator first step Sc1 are common, the first oscillator **1A**, the second oscillator **1B**, and the third oscillator **1C** of different types can be efficiently manufactured at low cost.

Second Embodiment

(112) FIG. **17** is a cross-sectional view of a common package according to a second embodiment.

(113) The common package **100** according to the embodiment is the same as the common package **100** of the first embodiment described above except that the circuit element **8** includes the temperature control element **7**. In the following description, the common package **100** of the embodiment is described with a focus on differences from the first embodiment described above, and a description of the similar matters is omitted. In the drawing of the embodiment, configurations similar to those of the above-described embodiment will be denoted by the same reference signs.

(114) As illustrated in FIG. **17**, in the common package **100** of the embodiment, the circuit element **8** includes the temperature control element **7**. In other words, the circuit element **8** has a function of the temperature control element **7**. Accordingly, the number of components is reduced, and manufacturing of the common package **100** is facilitated.

(115) As described above, in the embodiment, the circuit element **8** of the first oscillator **1A** includes the temperature control element **7**, and the circuit element **8** of the second oscillator **1B** includes the temperature control element **7**. Accordingly, the number of components is reduced, and manufacturing of the common package **100** is facilitated.

(116) According to the second embodiment, the same effects as in the first embodiment can also be exhibited.

Third Embodiment

(117) FIG. **18** is a cross-sectional view of a common package according to a third embodiment.

(118) The common package **100** according to the embodiment is the same as the common package **100** of the first embodiment described above except that the circuit element **4** includes the circuit element **8** and the temperature control element **7**. In the following description, the common package **100** of the embodiment is described with a focus on differences from the first embodiment described above, and a description of the similar matters is omitted. In the drawing of the embodiment, configurations similar to those of the above-described embodiment will be denoted by the same reference signs.

(119) As illustrated in FIG. **18**, in the common package **100** of the embodiment, the circuit element **4** includes the circuit element **8** and the temperature control element **7**. In other words, the circuit element **4** has functions of the circuit element **8** and the temperature control element **7**.

Accordingly, the number of components is reduced, and manufacturing of the common package **100** is facilitated. In the illustrated embodiment, the circuit element **4** is placed at an upper surface

of the inner package **3** in order to facilitate transfer of heat generated in the circuit element **4** for heating the resonator element **6** to the resonator element **6**.

(120) According to the third embodiment, the same effects as in the first embodiment described above can be exhibited.

(121) Although a method for manufacturing an oscillator according to the present disclosure has been described above based on the illustrated embodiments, the present disclosure is not limited thereto, and a configuration of each part can be replaced with any configuration having a similar function. Any other components may be added to the present disclosure. The embodiments may be combined as appropriate.

Claims

1. A method for manufacturing oscillators of a plurality of types including a first oscillator and a second oscillator, the method comprising: preparing a first container having a first outer base and a first lid, the first outer base having a first monolithic structure, the first outer base having a first surface and a second surface outwardly opposite to each other along a first direction; forming a first recess and a second recess in the first and second surfaces, respectively; manufacturing the first oscillator, the manufacturing the first oscillator including: a first-oscillator first step of mounting a first resonator element and a first circuit element in the first recess of the first outer base, the first circuit element being configured to oscillate the first resonator element to generate a first oscillation signal; and a first-oscillator second step of mounting a second resonator element in the second recess of the first outer base, an oscillation frequency of the second resonator element being controlled based on the first oscillation signal; preparing a second container having a second outer base and a second lid, the second outer base having a second monolithic structure, the second outer base having a third surface and a fourth surface outwardly opposite to each other along the first direction, wherein the second container has a same size and a same outer shape as the first container such that the first outer base has a same size and a same outer shape as the second outer base; forming a third recess and a fourth recess in the third and fourth surfaces, respectively; and manufacturing the second oscillator, the manufacturing the second oscillator including: a second-oscillator first step of mounting a third resonator element and a second circuit element in the third recess of the second outer base, the third resonator element having a same size as the first resonator element, the second circuit element having a same size as the first circuit element, the second circuit element being configured to oscillate the third resonator element to generate a second oscillation signal; and a second-oscillator second step of mounting a fourth resonator element in the fourth recess of the second outer base, an oscillation frequency of the fourth resonator element being controlled based on the second oscillation signal, the oscillation frequency of the fourth resonator element being different from the oscillation frequency of the second resonator element.
2. The method for manufacturing the oscillators according to claim 1, wherein in the first-oscillator first step, a first heater circuit configured to heat the first resonator element is further mounted in the first recess of the first container, and in the second-oscillator first step, a second heater circuit, which is a same type as the first heater circuit, configured to heat the third resonator element is further mounted in the third recess of the second container.
3. The method for manufacturing the oscillators according to claim 2, wherein the first circuit element includes the first heater circuit, and the second circuit element includes the second heater circuit.
4. The method for manufacturing the oscillators according to claim 1, wherein each of the first resonator element and the third resonator element is an SC-cut quartz crystal resonator element.
5. The method for manufacturing the oscillators according to claim 1, wherein each of the first resonator element and the third resonator element is an AT-cut quartz crystal resonator element.
6. The method for manufacturing the oscillators according to claim 1, wherein each of the second

resonator element and the fourth resonator element is an AT-cut quartz crystal resonator element.

7. The method for manufacturing the oscillators according to claim 1, wherein in the first-oscillator second step, a third circuit element configured to oscillate the second resonator element to generate a third oscillation signal is mounted in the second recess of the first container, and a frequency of the third oscillation signal is controlled based on the first oscillation signal, and in the second-oscillator second step, a fourth circuit element configured to oscillate the fourth resonator element to generate a fourth oscillation signal is further mounted in the fourth recess of the second container, and a frequency of the fourth oscillation signal is controlled based on the second oscillation signal.

8. The method for manufacturing the oscillators according to claim 7, wherein in the first-oscillator second step, the second resonator element and the third circuit element are mounted in the second recess of the first container in a state of being accommodated in a third container, and in the second-oscillator second step, the fourth resonator element and the fourth circuit element are mounted in the fourth recess of the second container in a state of being accommodated in a fourth container.

9. The method for manufacturing the oscillators according to claim 1, further comprising: manufacturing a third oscillator, the manufacturing the third oscillator including: a third-oscillator first step of mounting a fifth resonator element and a fifth circuit element to a fifth container having a same type as the first container, the fifth resonator element having a same size as the first resonator element, the fifth circuit element having a same size as the first circuit element and being configured to oscillate the fifth resonator element to generate a fifth oscillation signal.
